5G RAN

Radio Security Feature Parameter Description

Issue Draft A

Date 2025-12-31

HUAWEI TECHNOLOGIES CO., LTD.

Trademarks and Permissions

Notice

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: https://www.huawei.com

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added defense against DDoS attacks triggered by SRB scheduling feedback exceptions. For details, see: • **8.1 Principles** • **8.2.2 Impacts** • **8.4.1.1 Data Preparation** • **8.4.1.2 Using MML Commands**	Added the **SRB_SCH_EXP_SW** option to the **gNBAirIntfSecParam**.***AirIntfAntiDdosSw*** parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Modified the method of calculating the ratio of emergency UEs in a cell for the function of "preemption ratio restriction for multi-UE DDoS attacks over the air interface". For details, see **8.1 Principles**.	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010013	Radio Interface Ciphering	**4 Radio Interface Ciphering**
FBFD-021102	Integrity Protection	**5 Integrity Protection** **7 PDCP Counter Check**
FBFD-010023	Security Mechanism	**8 DDoS Attack Detection and Defense over the Air Interface**

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Radio Interface Ciphering	None	**4 Radio Interface Ciphering**
Integrity Protection	None	**5 Integrity Protection**

Function Name	Difference	Chapter/Section
PDCP Counter Check	Supported in both NSA and SA networking, with the following differences in what determines whether to release a DRB when PDCP counters are inconsistent: ● In NSA networking, a switch on the MeNB side determines whether to release the corresponding DRB. ● In SA networking, a switch on the gNodeB side determines whether to release the corresponding DRB.	**7 PDCP Counter Check**
DDoS attack detection and defense over the air interface	Supported in both NSA and SA networking, with the following differences: ● In NSA networking, this function is implemented on the eNodeB side. ● In SA networking, this function is implemented on the gNodeB side.	**8 DDoS Attack Detection and Defense over the Air Interface**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Radio Interface Ciphering	None	**4 Radio Interface Ciphering**
Integrity Protection	None	**5 Integrity Protection**
PDCP Counter Check	None	**7 PDCP Counter Check**

Function Name	Difference	Chapter/Section
DDoS attack detection and defense over the air interface	None	**8 DDoS Attack Detection and Defense over the Air Interface**

3 Overview

3.1 Introduction

Radio security refers to the confidentiality and integrity of data transmitted over the radio interface between gNodeBs and UEs. This document addresses two aspects of radio security:

- Radio interface ciphering: The sender uses a ciphering algorithm to convert plaintext data into ciphertext data and then sends the ciphertext data to the receiver, preventing information disclosure.
- Integrity protection: The sender uses an integrity protection algorithm to calculate a message authentication code for integrity (MAC-I) for a message and then sends the message and MAC-I to the receiver that will use the same integrity protection algorithm to calculate the X-MAC (expected MAC-I) for the received message. The receiver then compares the MAC-I with the X-MAC to ensure that the data has not been tampered with.

In NSA networking with the evolved packet core (EPC) as the core network, the gNodeB serves as the secondary gNodeB (SgNB). Between a UE and the SgNB, the control plane supports ciphering and integrity protection, and the user plane supports ciphering. If the gNodeB receives a user-plane security policy indication (Security Indication) from the master eNodeB (MeNB), then the support for user-plane integrity protection is determined by this indication. Control-plane data is carried on signaling radio bearer 3 (SRB3).

In SA networking, both the control plane and user plane between a UE and the gNodeB support ciphering and integrity protection.

NOTE

- Parameters related to air interface ciphering and integrity protection are not differentiated between the user plane and the control plane.
- The **UpIntegrityProtectionSw** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter on the LTE side needs to be selected to support user-plane integrity protection in NSA networking.
- For details about NSA networking, see *EPC-based NSA Basics*.
- For details about user-plane integrity protection activation by the SgNB in NSA networking, see section E.3.2 "Addition and modification of DRBs and/or SRB in SgNB" in 3GPP TS 33.401 V17.2.0.

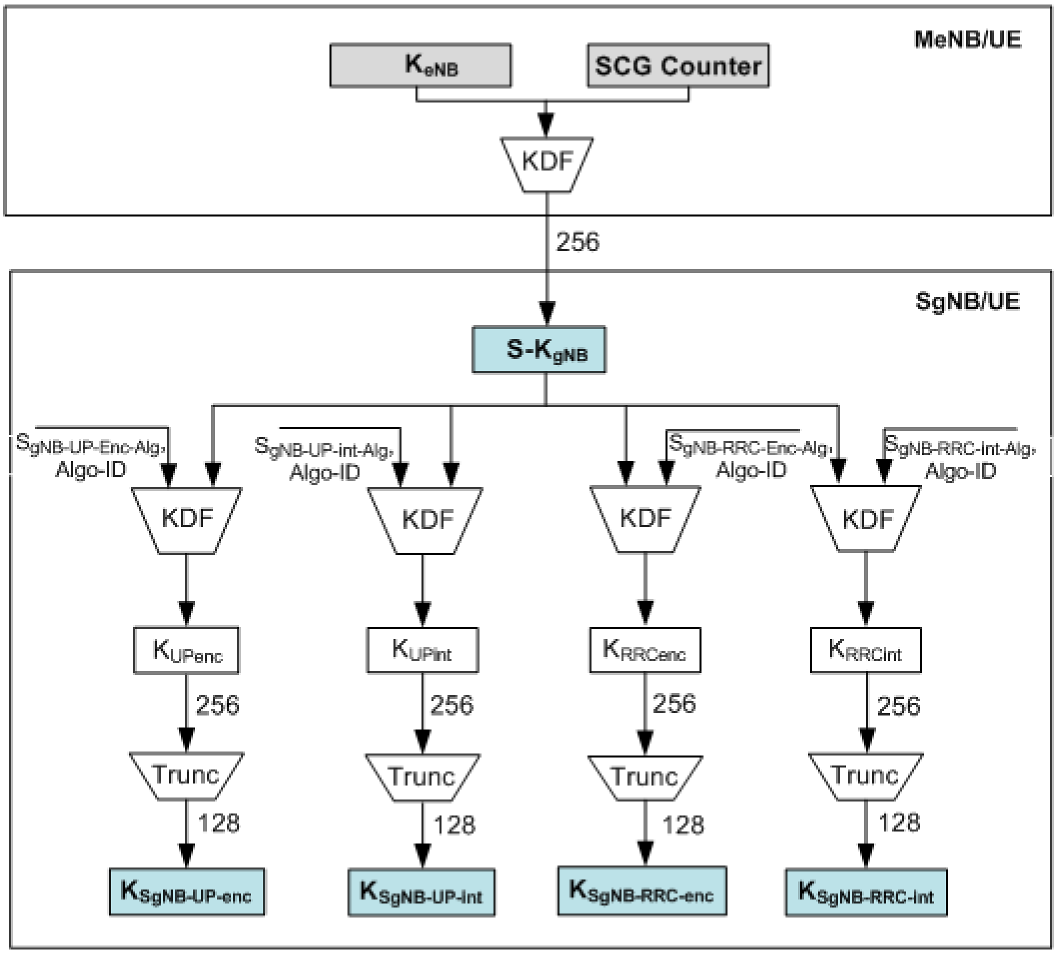
3.2 Key Derivation

gNodeBs perform message ciphering and integrity protection at the Packet Data Convergence Protocol (PDCP) layer. Cipher keys and integrity keys are important inputs to ciphering and integrity protection algorithms. These keys may be intercepted if they are transmitted over the radio interface. However, disclosure can be prevented if they are separately derived by UEs and gNodeBs. For details about the PDCP layer, see 3GPP TS 38.323 V1.0.0.

NSA Networking

Figure 3-1 illustrates the key derivation of the SgNB.

Figure 3-1 SgNB key derivation relationships



The master eNodeB (MeNB) sends a new secondary cell group (SCG) counter to the UE during each DC setup. The UE and MeNB respectively derive the S-K_{gNB} of the target SgNB based on the K_{eNB} and SCG counter. The MeNB sends the derived S-K_{gNB} to the SgNB. Both the UE and the SgNB derive the following keys:

- $K_{SgNB-UP-enc}$: used for user-plane ciphering by the SgNB
- $K_{SgNB-UP-int}$: used for user-plane integrity protection by the SgNB
- $K_{SgNB-RRC-enc}$: used for RRC signaling ciphering by the SgNB
- $K_{SgNB-RRC-int}$: used for RRC signaling integrity protection by the SgNB

In NSA networking, the SgNB control plane and user plane support ciphering and integrity protection. Therefore, $K_{SgNB-UP-int}$, $K_{SgNB-UP-enc}$, $K_{SgNB-RRC-int}$, and $K_{SgNB-RRC-enc}$ can be derived.

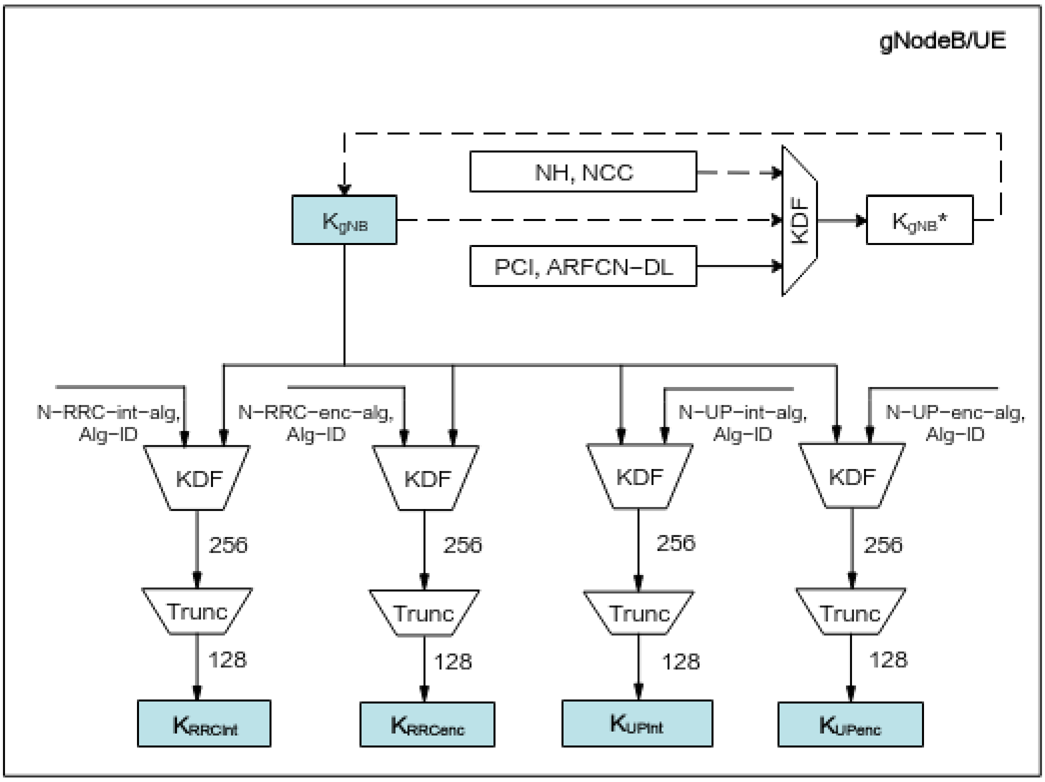
NOTE

For details about key derivation, see E.3.4 "Derivation of keys for RBs with PDCP in the SgNB" in 3GPP TS 33.401 V15.4.0.

SA Networking

Figure 3-2 illustrates the key derivation of the gNodeB.

Figure 3-2 gNodeB key derivation relationships



The next hop (NH) is derived by the UE and access and mobility management function (AMF) from the K_{AMF} and the last NH used. The next hop chaining count (NCC) is an NH counter and it counts the number of NH key chaining derivations. If keys need to be updated during a handover or state transition, NCC is used to synchronize key chaining between the UE and the gNodeB. Through this method, NCC helps determine whether the next K_{gNB}* is derived from the current K_{gNB} or from a new NH.

K_{gNB} is derived by the UE and AMF from K_{AMF}. During initial access stratum (AS) security context setup, K_{gNB} is used to derive K_{RRCint}, K_{RRCenc}, K_{UPint}, and K_{UPenc}. K_{gNB}* is a key derived by the UE and source gNodeB in a handover according to K_{gNB} (or a new NH), the target physical cell identifier (PCI), and the target downlink frequency. During a handover, the UE and target gNodeB use K_{gNB}* as the new K_{gNB} to derive the new K_{RRCint}, K_{RRCenc}, K_{UPint}, and K_{UPenc}.

The UE and gNodeB derive the following keys from K_{gNB}:

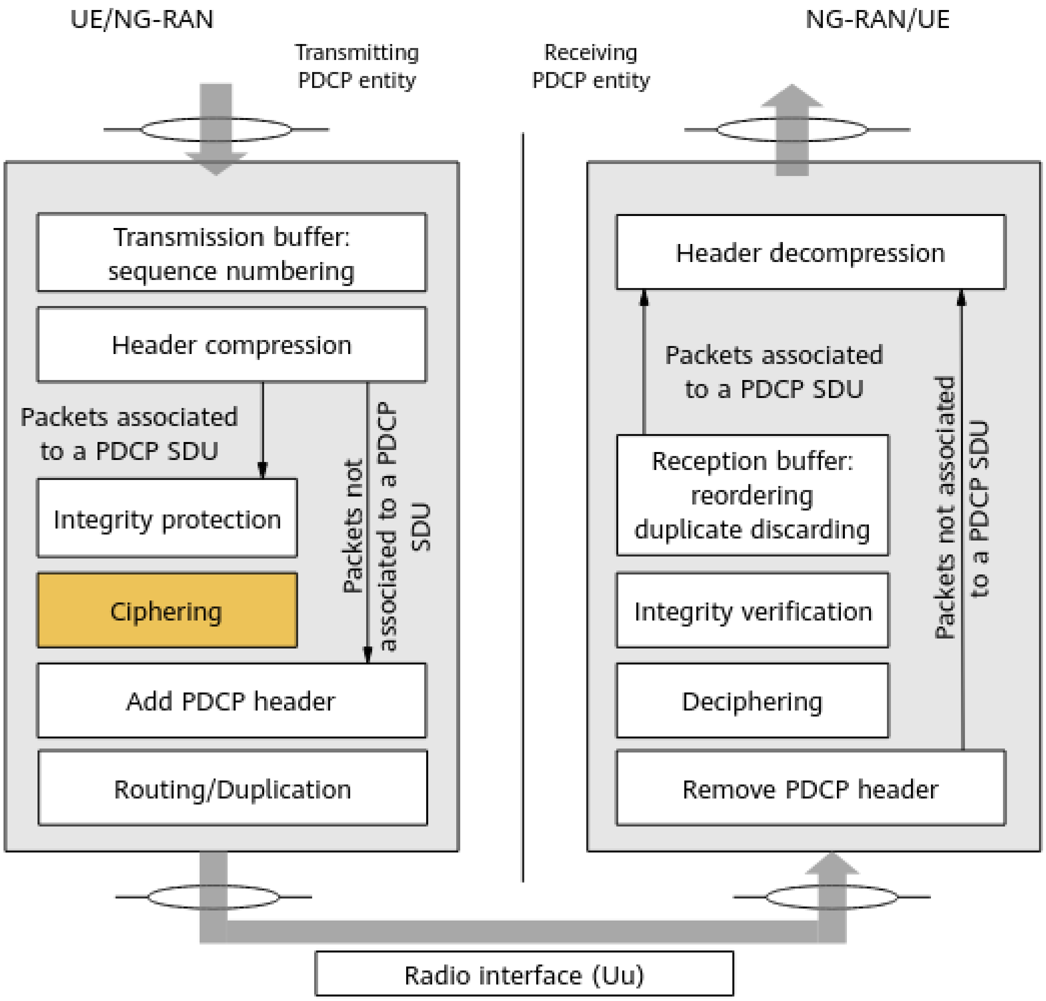
- K_{RRCint}: used for RRC signaling integrity protection by the gNodeB
- K_{RRCenc}: used for RRC signaling ciphering by the gNodeB
- K_{UPint}: used for user-plane integrity protection by the gNodeB
- K_{UPenc}: used for user-plane ciphering by the gNodeB

4 Radio Interface Ciphering

4.1 Principles

In radio interface ciphering, the sender and receiver negotiate a ciphering algorithm using RRC messages. The sender uses the negotiated ciphering algorithm to cipher messages and sends the ciphered messages to the receiver. The receiver uses the negotiated ciphering algorithm to decipher the received messages. gNodeBs cipher messages at the PDCP layer, as shown in **Figure 4-1**.

Figure 4-1 Ciphering in PDCP entities



Radio interface ciphering prevents the data between a gNodeB and UEs from being intercepted or disclosed. The following parameters can be configured on the gNodeB to determine the ciphering algorithms and priorities supported by the gNodeB. The configured ciphering algorithms and priorities are valid for all cells served by the gNodeB.

- The **gNBCipherCapb.*CipherAlgoPriority*** parameter specifies the priority of a ciphering algorithm. This parameter can be set to **PRIMARY**, **SECOND**, **THIRD**, or **FOURTH**. **Table 4-1** describes the values of ciphering algorithm priorities.

Table 4-1 Description of ciphering algorithm priorities

Ciphering Algorithm Priority	Description
PRIMARY	This value indicates that the ciphering algorithm has the highest priority.
SECOND	This value indicates that the ciphering algorithm has the second priority.

Ciphering Algorithm Priority	Description
THIRD	This value indicates that the ciphering algorithm has the third priority.
FOURTH	This value indicates that the ciphering algorithm has the fourth priority.

- The **gNBCipherCapb.*CipherAlgo*** parameter specifies the ciphering algorithm. This parameter can be set to **NULL**, **NOT_CONFIG**, **AES_128**, **SNOW3G_128**, or **ZUC_128**. **Table 4-2** describes ciphering algorithm values.

Table 4-2 Description of ciphering algorithm values

Ciphering Algorithm Value	Description
NULL	This value indicates that ciphering is not applied.
NOT_CONFIG	This value indicates that the gNodeB ignores the corresponding priority when selecting a ciphering algorithm.
AES_128	This value indicates the AES_128 algorithm.
SNOW3G_128	This value indicates the SNOW 3G_128 algorithm.
ZUC_128	This value indicates the ZUC_128 algorithm.

Radio interface ciphering requires that the involved gNodeB and UE use the same ciphering algorithm. **Table 4-3** lists ciphering algorithms and the corresponding IDs specified in 3GPP specifications. As specified in section 5.3.2 "User data and signalling data confidentiality" of 3GPP TS 33.501 V15.1.0, the gNodeB and UE must support the NEA0, NEA1, NEA2, and NEA3 algorithms to cipher control-plane and user-plane data.

Table 4-3 Ciphering algorithms and the corresponding IDs specified in 3GPP specifications

Ciphering Algorithm	Algorithm ID in 3GPP Specifications
NULL	NEA0
SNOW 3G_128	NEA1
AES_128	NEA2
ZUC_128	NEA3

NOTE

- For details about how to activate ciphering, see **6 Activation and Change of the Security Mode**.
- For details about ciphering on the radio interface, see chapter 5 "Security requirements and features" in 3GPP TS 33.501 V15.1.0.

4.2 Network Analysis

4.2.1 Benefits

The Radio Interface Ciphering feature prevents the user-plane data between a gNodeB and UEs from being intercepted or disclosed.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function is enabled by default and does not need to be activated. **Table 4-4** describes the parameters used for function optimization.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Ciphering Algorithm Priority	**gNBCipherCapb.*CipherAlgoPriority***	For details about the parameter setting, see **4.1 Principles**.
Ciphering Algorithm	**gNBCipherCapb.*CipherAlgo***	For details about the parameter setting, see **4.1 Principles**.

NOTE

It is recommended that the **gNBCipherCapb.*CipherAlgo*** parameter not be set to **NULL** or **NOT_CONFIG**. If this parameter is set to **NULL**, data is not ciphered. If this parameter is set to **NOT_CONFIG**, this priority is skipped during ciphering algorithm selection.

If the **gNBCipherCapb.*CipherAlgo*** parameter is set to **NULL** or the NULL ciphering algorithm has a higher priority (specified by **gNBCipherCapb.*CipherAlgoPriority***) than other ciphering algorithms, the gNodeB reports ALM-29901 gNodeB Radio Interface Ciphering Disabled, indicating that the radio interface ciphering configuration of the gNodeB is prone to information leakage risks.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

A gNodeB has been configured with ciphering algorithm priorities and corresponding ciphering algorithms by default, as listed in **Table 4-5**.

Table 4-5 gNodeB default ciphering algorithm priorities and ciphering algorithms

Ciphering Algorithm Priority	Ciphering Algorithm
PRIMARY	AES_128
SECOND	SNOW 3G_128
THIRD	ZUC_128
FOURTH	NOT_CONFIG

Optimization Command Examples

You can adjust the ciphering algorithm priorities and ciphering algorithms and set related parameters based on actual network scenarios.

```
//(Optional) Adjusting the ciphering algorithm priorities and ciphering algorithms
MOD GNBCIPHERCAPB: CipherAlgoPriority=PRIMARY,CipherAlgo=ZUC_128;
MOD GNBCIPHERCAPB: CipherAlgoPriority=SECOND,CipherAlgo=AES_128;
MOD GNBCIPHERCAPB: CipherAlgoPriority=THIRD,CipherAlgo=SNOW3G_128;
MOD GNBCIPHERCAPB: CipherAlgoPriority=FOURTH,CipherAlgo=NOT_CONFIG;
```

Deactivation Command Examples

If the **gNBCipherCapb.*CipherAlgo*** parameter is set to **NULL**, data is not ciphered.

```
//Adjusting the ciphering algorithm priorities and ciphering algorithms. In this example, the ciphering algorithm with the highest priority is set to NULL, and other ciphering algorithms are set to NOT_CONFIG.
MOD GNBCIPHERCAPB: CipherAlgoPriority=PRIMARY,CipherAlgo=NULL;
MOD GNBCIPHERCAPB: CipherAlgoPriority=SECOND,CipherAlgo=NOT_CONFIG;
MOD GNBCIPHERCAPB: CipherAlgoPriority=THIRD,CipherAlgo=NOT_CONFIG;
MOD GNBCIPHERCAPB: CipherAlgoPriority=FOURTH,CipherAlgo=NOT_CONFIG;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

NSA Networking

If the *sgNBtoMeNBContainer* IE in the Addition Request Acknowledge message of the SgNB traced over the X2 interface contains a ciphering algorithm, the Radio Interface Ciphering feature has taken effect. The *cipheringAlgorithm* IE carries the ciphering algorithm ID.

SA Networking

If the securityModeCommand message traced over the Uu interface contains a ciphering algorithm, the Radio Interface Ciphering feature has taken effect. The *cipheringAlgorithm* IE carries the ciphering algorithm ID.

The average number of UEs using different radio interface ciphering algorithms in a cell can be observed using the counters in **Table 4-6**.

Table 4-6 Counters related to radio interface ciphering algorithms

Counter ID	Counter Name
1911849109	**N.User.SecurAlg.NEA0.Avg**
1911849110	**N.User.SecurAlg.NEA1.Avg**
1911849111	**N.User.SecurAlg.NEA2.Avg**
1911849112	**N.User.SecurAlg.NEA3.Avg**

4.4.3 Network Monitoring

Table 4-7 lists the counters related to radio interface ciphering.

Table 4-7 Counters related to radio interface ciphering

Counter ID	Counter Name
1911825085	**N.SecurMode.SecAlgo.NEA0andNIA0**
1911825084	**N.SecurMode.CipherAlgo.NEA0**
1911825087	**N.SecureMode.UPCipher.Disable.RB**

4.4.4 Possible Issues

When the gNodeB reports ALM-29901 gNodeB Radio Interface Ciphering Disabled, information leakage risks exist under the current base station configuration. In this case, rectify the fault according to the alarm handling procedure.

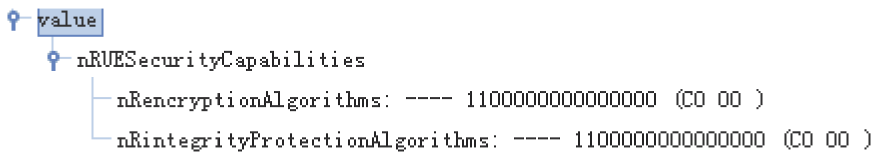
NSA Networking

After the initial configuration of the gNodeB is complete, an X2 message tracing task is started on the MAE-Access to trace messages over the X2 interface during NSA DC setup between the UE and the gNodeB. After the gNodeB sends an SgNB Addition Request Acknowledge message to the MeNB, if the gNodeB does not receive an SgNB Reconfiguration Complete message in time, the gNodeB sends an SgNB Release Required message to the MeNB to release the resource allocated to the UE.

Perform the following operations to identify and resolve the cause of the issue.

Step 1 Check UE security capabilities (including ciphering and integrity protection algorithms supported by the UE) by viewing the *nRUESecurityCapabilities* IE in the SgNB Addition Request message traced over the S1 interface, as shown in **Figure 4-2**.

Figure 4-2 *nRUESecurityCapabilities* IE in the SgNB Addition Request message



The *nRencryptionAlgorithms* IE indicates the ciphering algorithms supported by the UE and the *nRintegrityProtectionAlgorithms* IE indicates the integrity protection algorithms supported by the UE.

- If all of the bits are zero, the UE only supports the NULL algorithm.
- The leftmost bit of the IEs indicates whether the UE supports the SNOW 3G_128 algorithm.
- The second bit from the left indicates whether the UE supports the AES_128 algorithm.
- The third bit from the left indicates whether the UE supports the ZUC_128 algorithm.

Step 2 Check whether the gNodeB and UE support the same integrity protection and ciphering algorithms, based on the UE security capabilities and the configured algorithms on the gNodeB side.

- If they support the same integrity protection and ciphering algorithms, contact Huawei technical support.
- If they do not support the same integrity protection and ciphering algorithms, check the configuration of the UE's ciphering algorithms or contact the UE vendor.

----End

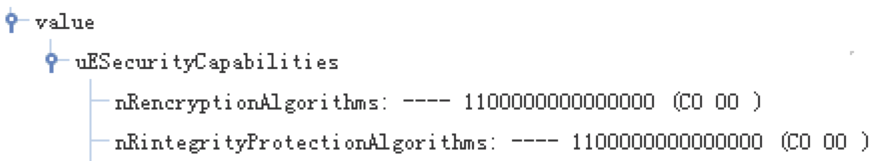
SA Networking

After the initial configuration of the gNodeB is complete, a Uu interface tracing task is started on the MAE-Access to trace Uu interface messages transmitted between the UE and the gNodeB during security mode setup. After the gNodeB sends an AS Security Mode Command message to the UE, if the gNodeB does not receive a Security Mode Complete message from the UE in time or receives a Security Mode Failure message from the UE, the gNodeB sends an RRC Connection Release message to the UE to release the resources allocated to the UE.

Perform the following operations to identify and resolve the cause of the issue.

Step 1 Check UE security capabilities (including ciphering and integrity protection algorithms supported by the UE) by viewing the *uESecurityCapabilities* IE in the INITIAL_CONTEXT_SETUP_REQ message traced over the NG interface, as shown in **Figure 4-3**.

Figure 4-3 *uESecurityCapabilities* IE in the INITIAL_CONTEXT_SETUP_REQ message



The *nRencryptionAlgorithms* IE indicates the ciphering algorithms supported by the UE and the *nRintegrityProtectionAlgorithms* IE indicates the integrity protection algorithms supported by the UE.

- If all of the bits are zero, the UE only supports the NULL algorithm.
- The leftmost bit of the IEs indicates whether the UE supports the SNOW 3G_128 algorithm.
- The second bit from the left indicates whether the UE supports the AES_128 algorithm.
- The third bit from the left indicates whether the UE supports the ZUC_128 algorithm.

Step 2 Check whether the gNodeB and UE support the same integrity protection and ciphering algorithms, based on the UE security capabilities and the configured algorithms on the gNodeB side.

- If they support the same integrity protection and ciphering algorithms, contact Huawei technical support.

- If they do not support the same integrity protection and ciphering algorithms, check the configuration of the UE's ciphering algorithms or contact the UE vendor.

----End

5 Integrity Protection

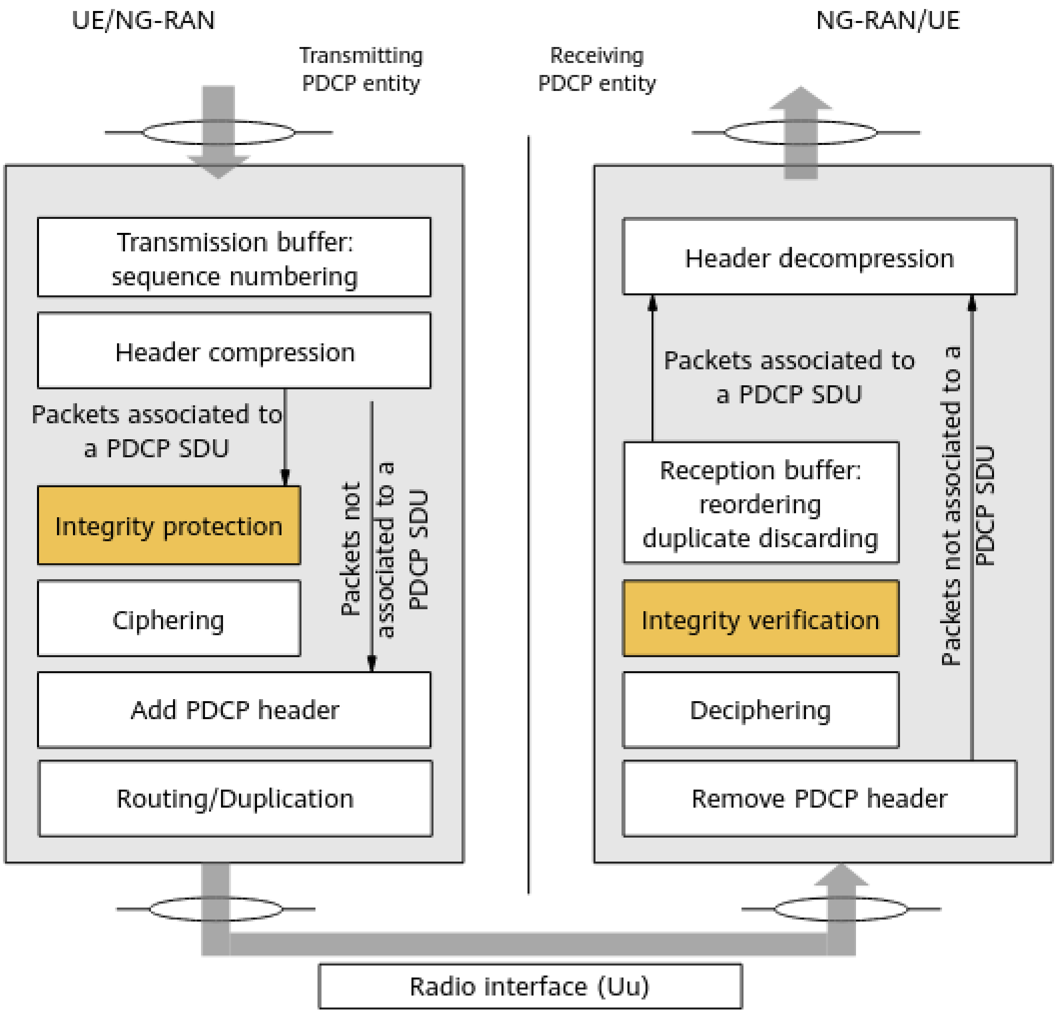
5.1 Principles

In integrity protection, the sender and receiver negotiate an integrity protection algorithm by exchanging RRC messages. The sender then uses the negotiated algorithm to calculate the MAC-I for a message. The sender sends the MAC-I and message to the receiver, and then the receiver uses the negotiated integrity protection algorithm to calculate the X-MAC of the message and compares it with the MAC-I in the received message.

- If two codes are different, the message has been tampered with.
- If two codes are the same, the message has not been tampered with and passed the integrity verification.

The gNodeB protects the integrity of messages at the PDCP layer, as shown in **Figure 5-1**.

Figure 5-1 Integrity protection in PDCP entities



Integrity protection enables receivers (either UEs or gNodeBs) to check whether messages have been tampered with. Integrity protection must be performed on all RRC signaling messages, except those that do not require integrity protection as listed in 3GPP TS 38.331. The following two parameters specify the integrity protection algorithms and priorities supported by a gNodeB. These parameters are valid for all cells served by the gNodeB.

- The **gNBIntegrityCapb.*IntegrityAlgoPriority*** parameter specifies the priority of an integrity protection algorithm. This parameter can be set to **PRIMARY**, **SECOND**, or **THIRD**. Table 5-1 describes the values of integrity protection algorithm priorities.

Table 5-1 Description of integrity protection algorithm priorities

Integrity Protection Algorithm Priority	Description
PRIMARY	This value indicates that the integrity protection algorithm has the highest priority.

Integrity Protection Algorithm Priority	Description
SECOND	This value indicates that the integrity protection algorithm has the second priority.
THIRD	This value indicates that the integrity protection algorithm has the third priority.

- The **gNBIntegrityCapb.*IntegrityAlgo*** parameter specifies an integrity protection algorithm. This parameter can be set to **NOT_CONFIG**, **AES_128**, **SNOW3G_128**, or **ZUC_128**. **Table 5-2** describes the values of integrity protection algorithms.

Table 5-2 Description of integrity protection algorithms

Integrity Protection Algorithm Value	Description
NOT_CONFIG	This value indicates that the gNodeB ignores the corresponding priority when selecting an integrity protection algorithm.
AES_128	This value indicates the AES_128 algorithm.
SNOW3G_128	This value indicates the SNOW 3G_128 algorithm.
ZUC_128	This value indicates the ZUC_128 algorithm.

Integrity protection requires that the involved gNodeB and UE use the same integrity protection algorithm. **Table 5-3** lists integrity protection algorithms and the corresponding IDs specified in 3GPP specifications. As specified in section 5.3.3 "User data and signalling data integrity" of 3GPP TS 33.501 V15.1.0, the gNodeB and UE must support the NIA1, NIA2, and NIA3 algorithms to protect integrity of control-plane and user-plane data.

Table 5-3 Integrity protection algorithms and the corresponding IDs specified in 3GPP specifications

Integrity Protection Algorithm	Algorithm ID in 3GPP Specifications
SNOW 3G_128	NIA1
AES_128	NIA2
ZUC_128	NIA3

NOTE

- For details about how to activate integrity protection, see **6 Activation and Change of the Security Mode**.
- For details about integrity protection on the radio interface, see chapter 5 "Security requirements and feature" in 3GPP TS 33.501 V15.1.0.

5.2 Network Analysis

5.2.1 Benefits

Integrity protection enables receiving entities (either UEs or gNodeBs) to check whether RRC signaling and user-plane data have been tampered with.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

This function is enabled by default and does not need to be activated. **Table 5-4** describes the parameters used for function optimization.

Table 5-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Integrity Protection Algorithm Priority	***gNBIntegrityCapb.IntegrityAlgoPriority***	For details about the parameter setting, see **5.1 Principles**.

Parameter Name	Parameter ID	Setting Notes
Integrity Protection Algorithm	**gNBIntegrityCapb.*IntegrityAlgo***	For details about the parameter setting, see **5.1 Principles**.

NOTE

It is recommended that the **gNBIntegrityCapb.*IntegrityAlgo*** parameter not be set to **NOT_CONFIG**. If this parameter is set to **NOT_CONFIG**, the priority is skipped during integrity protection algorithm selection.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

A gNodeB has been configured with integrity protection algorithm priorities and the corresponding integrity protection algorithms by default, as listed in **Table 5-5**.

Table 5-5 gNodeB default integrity protection algorithm priorities and the corresponding integrity protection algorithms

Integrity Protection Algorithm Priority	Integrity Protection Algorithm
PRIMARY	AES_128
SECOND	SNOW3G_128
THIRD	ZUC_128

Optimization Command Examples

```
//(Optional) Adjusting the integrity protection algorithm priority and the corresponding integrity protection algorithm
MOD GNBINTEGRITYCAPB: IntegrityAlgoPriority =PRIMARY, IntegrityAlgo =ZUC_128;
//(Optional) Adjusting the integrity protection algorithm priority and the corresponding integrity protection algorithm
MOD GNBINTEGRITYCAPB: IntegrityAlgoPriority =SECOND, IntegrityAlgo =AES_128;
//(Optional) Adjusting the integrity protection algorithm priority and the corresponding integrity protection
```

```
algorithm
MOD GNBINTEGRITYCAPB: IntegrityAlgoPriority =THIRD, IntegrityAlgo =SNOW3G_128;
```

Deactivation Command Examples

Integrity protection cannot be deactivated and at least one integrity protection algorithm must be configured.

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

NSA Networking

- When the gNodeB and UE support SRB3, if the *sgNBtoMeNBContainer* IE in the Addition Request Acknowledge message of the SgNB traced over the X2 interface contains an integrity protection algorithm, the integrity protection function has taken effect. The *integrityProtAlgorithm* IE in this message carries the integrity protection algorithm ID.
- When the gNodeB and UE do not support SRB3, the *sgNBtoMeNBContainer* IE in the Addition Request Acknowledge message of the SgNB traced over the X2 interface does not contain the integrity protection algorithm and the *integrityProtAlgorithm* IE.

SA Networking

If the securityModeCommand message traced over the Uu interface contains an integrity protection algorithm, the integrity protection function has taken effect. The *integrityProtAlgorithm* IE in this message carries the integrity protection algorithm ID.

The average number of UEs using different integrity protection algorithms in a cell can be observed using the counters in **Table 5-6**.

Table 5-6 Counters related to integrity protection algorithms

Counter ID	Counter Name
1911849113	**N.User.SecurAlg.NIA0.Avg**
1911849114	**N.User.SecurAlg.NIA1.Avg**
1911849115	**N.User.SecurAlg.NIA2.Avg**
1911849116	**N.User.SecurAlg.NIA3.Avg**

NOTE

According to the requirements of section E.3.10 "Protection of the traffic between the UE and SgNB" in 3GPP TS 33.401 V18.1.0 and section 5.3.3 "User data and signalling data integrity" in 3GPP TS 33.501 V18.6.0, the **N.User.SecurAlg.NIA0.Avg** counter in **Table 5-6** applies only to the SA Option 2 architecture.

5.4.3 Network Monitoring

Table 5-7 lists the counters related to integrity protection.

Table 5-7 Counters related to integrity protection

Counter ID	Counter Name
1911825085	**N.SecurMode.SecAlgo.NEA0andNIA0**
1911825086	**N.SecureMode.UPIntegrity.Disable.RB**
1911825088	**N.PDCP.UL.UPIntegrityCheck.Failure**
1911825089	**N.PDCP.UL.CPIntegrityCheck.Failure**

5.4.4 Possible Issues

The troubleshooting for integrity protection is the same as that for radio interface ciphering. For details, see **4.4.4 Possible Issues**.

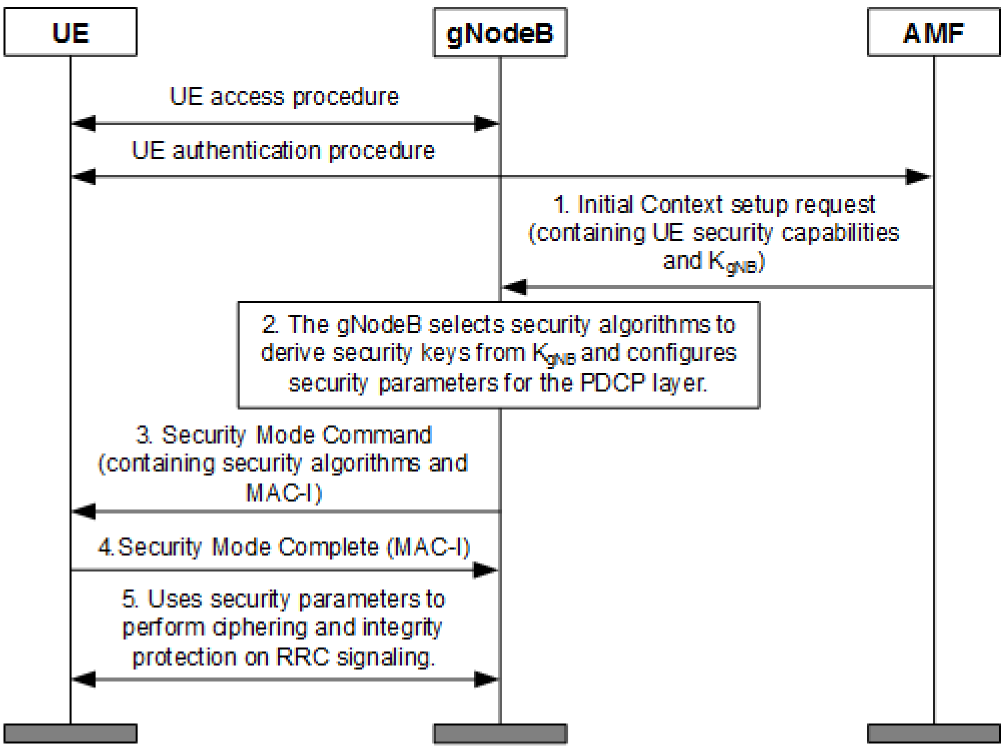
6 Activation and Change of the Security Mode

6.1 Initial Security Mode Activation Procedure

AS SMC Procedure

After receiving the UE security context from the AMF, the gNodeB triggers an AS security mode command (SMC) procedure, as shown in **Figure 6-1**. During this procedure, security algorithms are negotiated, and ciphering and integrity protection of RRC signaling are activated.

Figure 6-1 AS SMC procedure



1. After an RRC connection is set up between a UE and the gNodeB, the AMF generates the K_{gNB} and NH and sends the UE security capabilities and K_{gNB} to the gNodeB. The UE security capabilities include the ciphering and integrity protection algorithms supported by the UE.
2. The gNodeB obtains the intersection of the list of prioritized ciphering algorithms, list of prioritized integrity protection algorithms, and UE security capabilities. The gNodeB then selects the highest-priority integrity protection algorithm and the highest-priority ciphering algorithm. The gNodeB uses K_{gNB} and the selected algorithms to derive K_{RRCenc} and K_{RRCint}, and configures related ciphering and integrity protection parameters for the PDCP layer.
3. The gNodeB sends the UE a Security Mode Command message, which contains the selected security algorithms. The message is sent through SRB1 and is integrity-protected but not ciphered by the gNodeB.
4. The gNodeB receives a response from the UE:
   - If the response is a Security Mode Complete message, the security mode activation succeeds. The Security Mode Complete message is sent through SRB1 and is integrity-protected but not ciphered by the UE.
   - If the response is a Security Mode Failure message, the security mode activation fails. The Security Mode Failure message is sent through SRB1 without ciphering or integrity protection.
5. If the security mode activation succeeds, ciphering and integrity protection of RRC signaling are activated.

NOTE

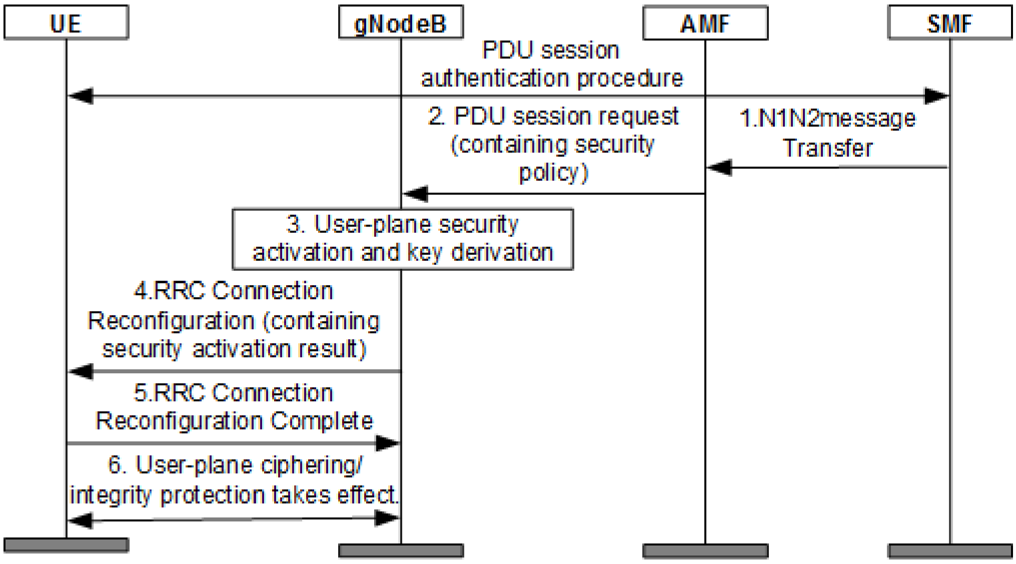
For the UE, the integrity protection and ciphering algorithms negotiated during the AS SMC procedure are applicable to all SRBs and DRBs.

For details about the initial security mode activation procedure, see section 6.7.4 "AS security mode command procedure" in 3GPP TS 33.501 V15.1.0 and section 5.3.4 "Initial security activation" in 3GPP TS 38.331 V15.2.0.

User-Plane Security Activation Procedure

After receiving the user-plane security policy message from the session management function (SMF), the gNodeB triggers a user-plane security activation procedure, as shown in **Figure 6-2**. In this procedure, the user-plane DRB is established, and ciphering and integrity protection of the user-plane data over the DRB are activated.

Figure 6-2 User-plane security activation procedure



1. After the AS SMC procedure is complete, the SMF determines the security policy on the user-plane data in the PDU session and sends the security policy to the AMF through an N1N2messageTransfer message during PDU session setup.
2. The AMF sends the security policy to the gNodeB through a PDU session request message. The security policy contains the effective indication of ciphering and integrity protection.
3. The gNodeB determines whether to activate the ciphering and integrity protection of the user-plane data in the PDU session based on the security policy delivered by the SMF and the security capabilities of the gNodeB.
   - If the gNodeB decides to activate user-plane ciphering, the gNodeB needs to calculate the K_{UPenc}.
   - If the gNodeB decides to activate user-plane integrity protection, the gNodeB needs to calculate the K_{UPint} and configure the corresponding ciphering and integrity protection parameters for the PDCP layer.

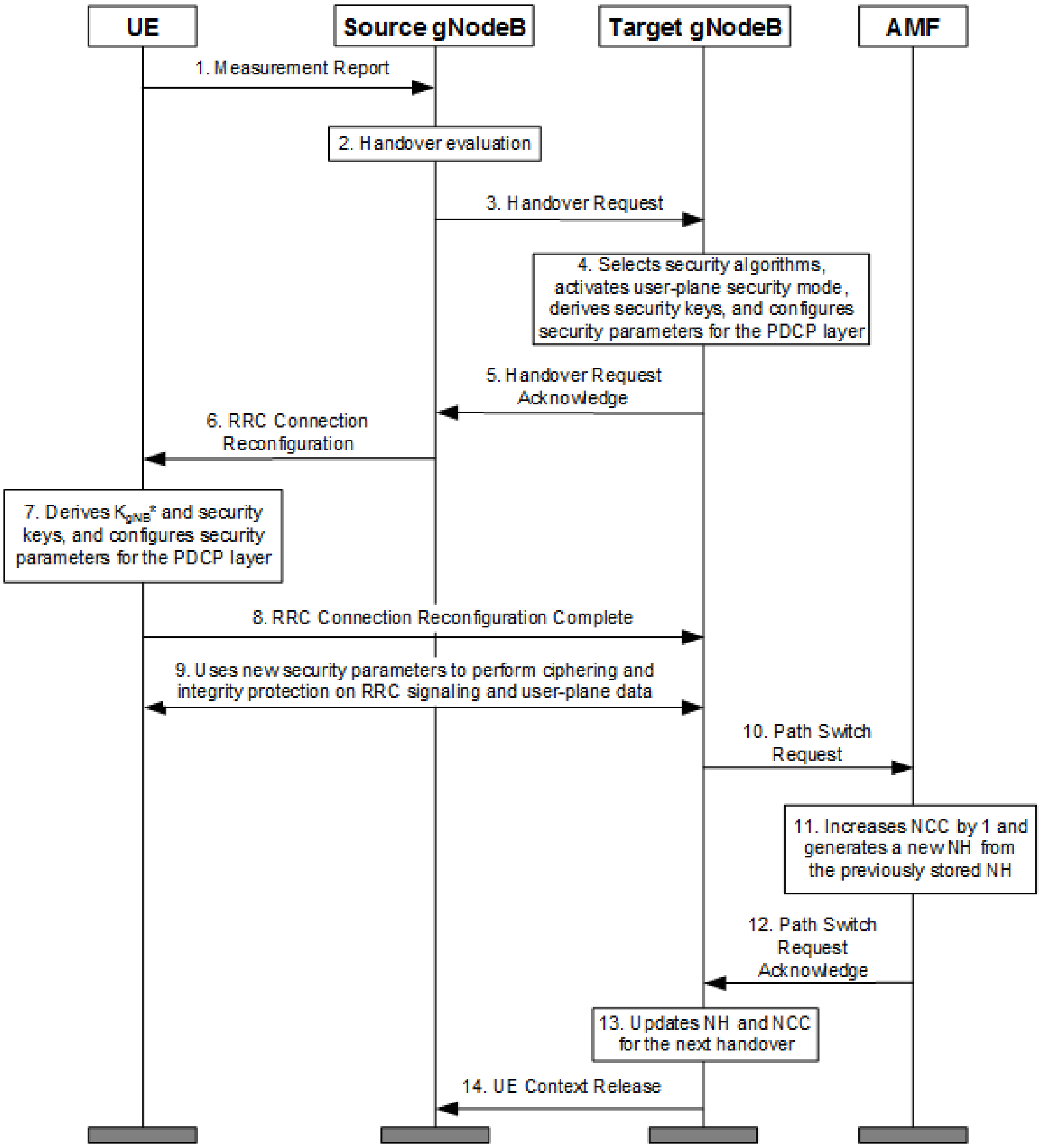
   The method of calculating the keys is as follows: The gNodeB calculates K_{UPenc} and K_{UPint} based on K_{gNB} and the ciphering and integrity protection algorithms selected in the AS SMC procedure.
4. The gNodeB sends an RRC Connection Reconfiguration message to the UE, informing the UE of the effective indication of user-plane ciphering and integrity protection.
5. The gNodeB receives a response from the UE:
   - If the gNodeB receives an RRC Connection Reconfiguration Complete message from the UE, the user-plane security activation is successful.
   - If the gNodeB receives an RRC Connection Reconfiguration Failure message from the UE, the user-plane security activation fails.
6. If the user-plane security activation is successful, both ciphering and integrity protection are activated on the user plane.

For details on the user-plane security activation procedure, see section 6.6.2 "UP security activation mechanism" in 3GPP TS 33.501 V15.1.0.

6.2 Security Handling Procedure During Handovers

A handover may be performed over the Xn or NG interface. A handover over the Xn interface is called Xn handover, and that over the NG interface is called NG handover. During handovers, ciphering and integrity protection algorithms may change. During a handover or an RRC state transition, the keys used for ciphering and integrity protection may also change. The security handling procedure for Xn handovers is similar to that for NG handovers. This section describes the security handling procedure for Xn handovers, as shown in **Figure 6-3**.

Figure 6-3 Security handling procedure during an Xn handover



The security handling procedure during an Xn handover is as follows:

1. The UE sends a Measurement Report message to the source gNodeB.

2. The source gNodeB decides to perform an Xn handover for the UE.
   – If the NH of the source gNodeB is not used, the NH, the target PCI, and the target downlink frequency are used to generate the K_{gNB}*.
   – If the NH of the source gNodeB is already used, the K_{gNB}, the target PCI, and the target downlink frequency are used to generate the K_{gNB}*.
3. The source gNodeB encapsulates the security context (including the UE security capabilities, user-plane security policy, NCC, and K_{gNB}*) into a Handover Request message and sends the message to the target gNodeB over the X2 interface.
4. After receiving the Handover Request message, the target gNodeB performs the following actions:
   a. It selects the highest-priority integrity protection and ciphering algorithms supported by the UE, based on the local prioritized algorithms.
   b. It activates the user-plane security based on the locally configured ciphering and integrity protection capabilities and the user-plane security policy.
   c. It uses K_{gNB}* forwarded by the source gNodeB as K_{gNB}, and uses K_{gNB} and the selected security algorithms to derive ciphering and integrity protection keys for RRC signaling and user-plane data.
   d. It configures post-handover security parameters for the PDCP layer based on the selected ciphering and integrity protection algorithms and keys. The security parameters will be used after the handover.
5. The target gNodeB forwards the NCC, user-plane security effectiveness result, and the selected security algorithms to the source gNodeB through a Handover Request Acknowledge message.
6. The source gNodeB sends an RRC Connection Reconfiguration message containing the NCC, user-plane security effectiveness result, and security algorithms provided by the target gNodeB to the UE. The RRC Connection Reconfiguration message is ciphered and integrity-protected by using the pre-handover security-related parameters.
7. The UE generates K_{gNB}* based on its own K_{gNB} and the received NCC, and derives ciphering and integrity protection keys for RRC signaling and user-plane data based on the user-plane security effectiveness result, security algorithms, and K_{gNB}*. The UE then configures post-handover security parameters for the PDCP layer based on the ciphering and integrity algorithms selected by the UE.
8. After the UE is handed over, it sends an RRC Connection Reconfiguration Complete message to the target gNodeB. The message is ciphered and integrity-protected using the post-handover security parameters.
9. The target gNodeB uses the post-handover security parameters to perform ciphering and integrity protection on RRC signaling and user-plane data.
10. The target gNodeB sends a Path Switch Request message to the AMF, informing the AMF that the UE is handed over.
11. After receiving the Path Switch Request message, the AMF increases NCC by 1 and derives a new NH from the one that was previously stored.
12. The AMF sends a Path Switch Request Acknowledge message to the target gNodeB. This message contains the new NCC and NH.

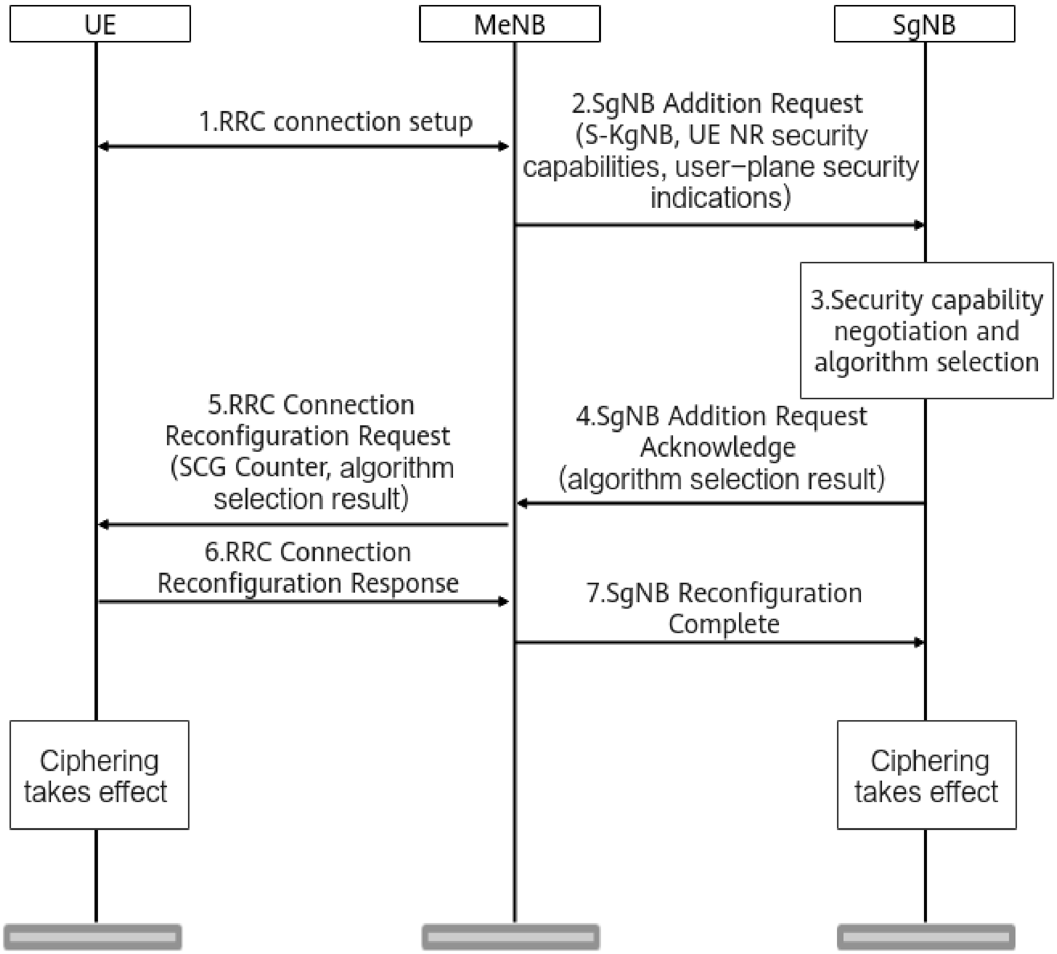
13. The target gNodeB saves the new NCC and NH for the next handover. If the target gNodeB does not receive a Path Switch Request Acknowledge message, the NCC and NH at the target gNodeB are not updated. In such a case, the target gNodeB must use K_{gNB} to derive K_{gNB}* in the next handover.
14. The target gNodeB sends a UE Context Release message to the source gNodeB, instructing the source gNodeB to release the UE context.

For details about the security handling procedure during handovers, see chapter 6 "Security procedures between UE and 5G network functions" in 3GPP TS 33.501 V15.1.0.

6.3 SgNB Security Mode Activation Procedure During an NSA DC Setup

Figure 6-4 shows the SgNB security mode activation procedure during an NSA DC setup.

Figure 6-4 SgNB security mode activation procedure during an NSA DC setup



1. The UE and the MeNB establish an RRC connection.
2. Before the MeNB decides to add an SgNB for DC, the MeNB checks whether the UE has the permission to access New Radio (NR). If the UE has NR access

permission, the MeNB sends information such as the UE NR security capabilities, S-K_{gNB}, and user-plane security indications to the target SgNB through an SgNB Addition Request message.

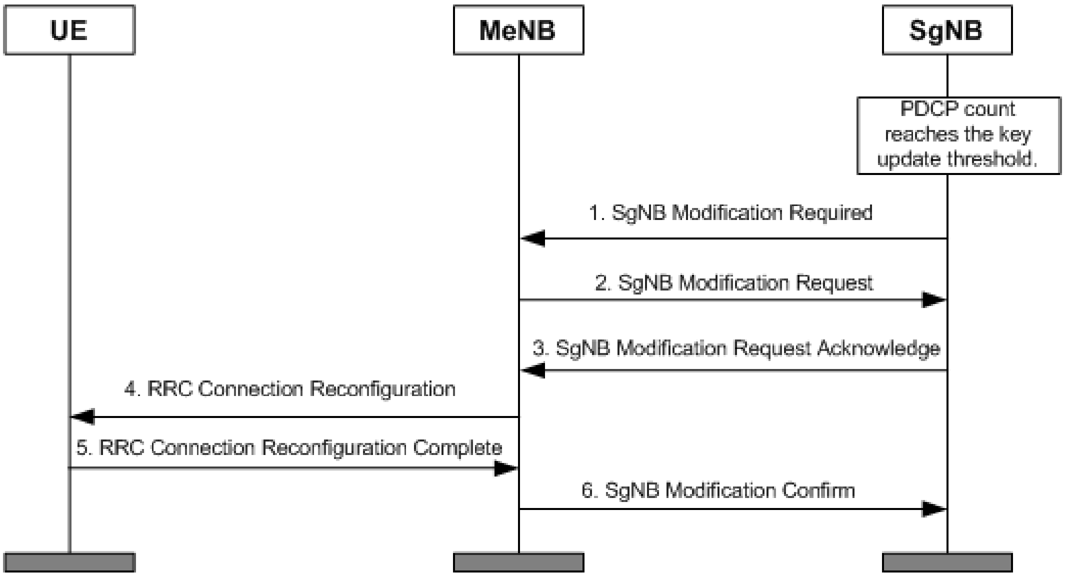
3. After receiving an SgNB Addition Request message from the MeNB, the SgNB performs the following operations:
   a. The SgNB selects the highest-priority ciphering algorithm supported by the UE, based on the local prioritized ciphering algorithms.
   b. The SgNB derives $K_{SgNB-UP-enc}$, $K_{SgNB-RRC-int}$, and $K_{SgNB-RRC-enc}$ based on S-K_{gNB} forwarded by the MeNB and the selected security algorithms. If user-plane integrity protection needs to be activated according to the user-plane security indication, $K_{SgNB-UP-int}$ needs to be derived.
   c. The SgNB configures security parameters for the PDCP layer according to the selected ciphering algorithm and key.
4. The SgNB sends the algorithm selection result to the MeNB through an SgNB Addition Request Acknowledge message.
5. The MeNB sends the SgNB security algorithm selection result and the SCG counter to the UE through an RRC Connection Reconfiguration Request message.
6. The UE calculates the S-K_{gNB} based on the received SCG counter, and then calculates the corresponding $K_{SgNB-UP-enc}$, $K_{SgNB-RRC-int}$, and $K_{SgNB-RRC-enc}$ based on the S-K_{gNB} and the received security algorithm selection result. If user-plane integrity protection needs to be activated according to the user-plane security indication, the corresponding $K_{SgNB-UP-int}$ needs to be calculated. The UE sends an RRC Connection Reconfiguration Response message to the MeNB.
7. The MeNB sends an SgNB Reconfiguration Complete message to the SgNB to notify the SgNB that the configuration is complete. After receiving this message, the ciphering function takes effect between the SgNB and the UE.

6.4 Key Update Process

6.4.1 S-K_{gNB} Update Process Triggered by the SgNB

When detecting that the uplink or downlink PDCP count for a DRB is about to wrap around, the SgNB triggers an S-K_{gNB} update process, as shown in **Figure 6-5**.

Figure 6-5 S-K_{gNB} update process triggered by the SgNB



1. The SgNB sends an SgNB Modification Required message to the MeNB. This message carries the *PDCP Change Indication* IE, indicating that the S-K_{gNB} needs to be updated.

   NOTE

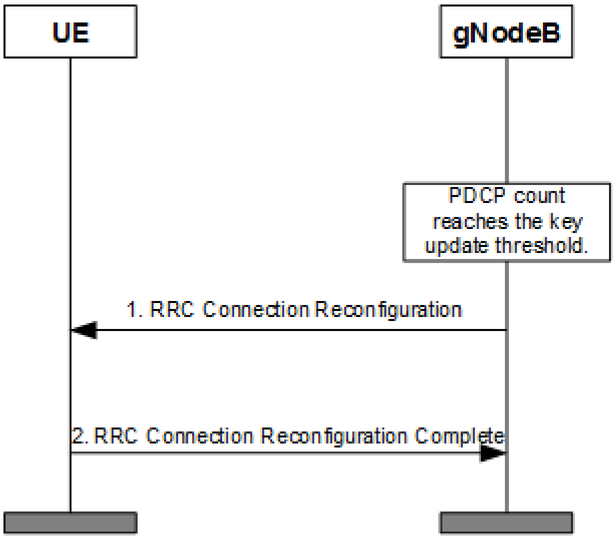
   For details about the *PDCP Change Indication* IE, see section 9.2.109 "PDCP Change Indication" in 3GPP TS 36.423 V15.1.0.

2. Upon receiving the SgNB Modification Required message, the MeNB automatically increases the SCG counter by 1 and calculates a new S-K_{gNB}. The MeNB then sends an SgNB Modification Request message to the SgNB, carrying the new S-K_{gNB}.
3. After receiving the new S-K_{gNB}, the SgNB derives new $K_{SgNB\text{-}UP\text{-}enc}$, $K_{SgNB\text{-}RRC\text{-}int}$, and $K_{SgNB\text{-}RRC\text{-}enc}$. If user-plane integrity protection needs to be activated according to the user-plane security indication, a new $K_{SgNB\text{-}UP\text{-}int}$ needs to be derived. The PDCP count is reset after the preceding keys are derived. The SgNB returns an SgNB Modification Request Acknowledge message to the MeNB.
4. The MeNB sends an RRC Connection Reconfiguration message, carrying the new SCG counter, to the UE.
5. After receiving the new SCG counter, the UE calculates a new S-K_{gNB} and new $K_{SgNB\text{-}UP\text{-}enc}$, $K_{SgNB\text{-}RRC\text{-}int}$, and $K_{SgNB\text{-}RRC\text{-}enc}$. If user-plane integrity protection needs to be activated according to the user-plane security indication, a new $K_{SgNB\text{-}UP\text{-}int}$ needs to be calculated. After calculating the preceding keys, the UE returns an RRC Connection Reconfiguration Complete message to the MeNB.
6. The MeNB sends an SgNB Modification Confirm message to the SgNB. After this message is received, the new $K_{SgNB\text{-}UP\text{-}enc}$, $K_{SgNB\text{-}RRC\text{-}int}$, and $K_{SgNB\text{-}RRC\text{-}enc}$ between the UE and the SgNB take effect. If user-plane integrity protection needs to be activated according to the user-plane security indication, the new $K_{SgNB\text{-}UP\text{-}int}$ between the UE and the SgNB takes effect.

6.4.2 AS Key Update Triggered by the gNodeB

When the gNodeB detects that the uplink or downlink PDCP count for an SRB or DRB is about to wrap around, an AS key update procedure is triggered, as shown in **Figure 6-6**.

Figure 6-6 AS key update triggered by the gNodeB



1. The gNodeB sends an RRC Connection Reconfiguration message to the UE to trigger an intra-cell handover procedure for a key update. The gNodeB derives a new K_{gNB}*, and uses this K_{gNB}* as K_{gNB} to derive K_{RRCint}, K_{RRCenc}, K_{UPint}, and K_{UPenc}.
2. Upon receiving the message from the gNodeB, the UE derives the same K_{gNB}*, and uses the K_{gNB}* as K_{gNB} to derive K_{RRCint}, K_{RRCenc}, K_{UPint}, and K_{UPenc}. The UE returns an RRC Connection Reconfiguration Complete message to the gNodeB.

6.5 Counters Related to Security Mode Activation and Change Procedures

Counter ID	Counter Name
1911817075	**N.SecurMode.Req**
1911817073	**N.SecurMode.Cmd**
1911817074	**N.SecurMode.Cmp**
1911817072	**N.SecurMode.Fail**
1911825085	**N.SecurMode.SecAlgo.NEA0andNIA0**
1911825084	**N.SecurMode.CipherAlgo.NEA0**

7 PDCP Counter Check

7.1 Principles

PDCP counters serve as the source data of the radio interface ciphering and integrity protection keys. If the PDCP counters between a UE and a base station are inconsistent, the keys calculated by the UE and the base station will also be inconsistent, affecting radio interface ciphering and deciphering. If integrity protection is not enabled on the user plane, then user-plane data is prone to attacks and risks such as packet loss. To solve this problem, 3GPP specifications separately define the PDCP counter check function for NSA (EN-DC) networking and SA networking. The PDCP counter check function requires a base station to proactively check user-plane PDCP counters. This function is controlled by the **gNBAirIntfSecParam.*CounterCheckSwitch*** parameter.

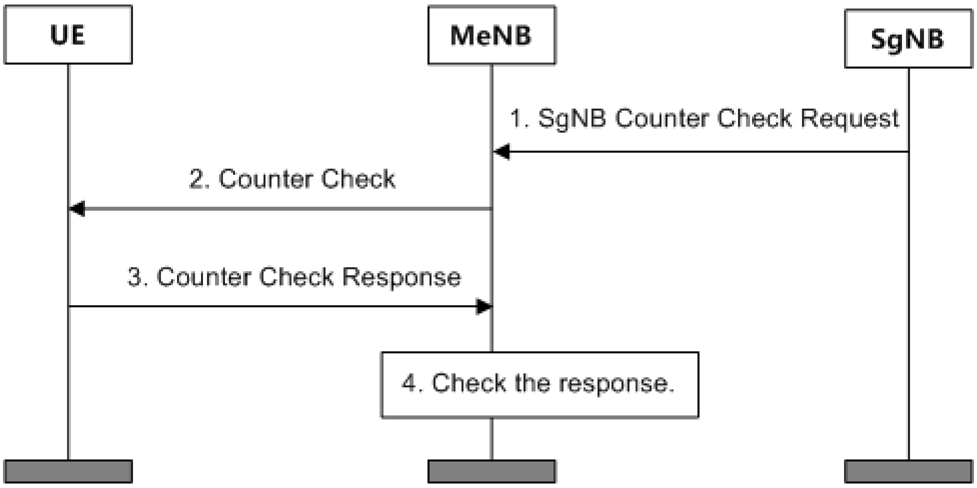
NOTE

- For details about the PDCP counter check in NSA (EN-DC) networking, see section E.3.7 "Periodic local authentication procedure" in 3GPP TS 33.401 V15.5.0.
- For details about the PDCP counter check in SA networking, see section 6.13 "Signalling procedure for PDCP COUNT check" in 3GPP TS 33.501 V15.4.0.

NSA (EN-DC) Networking

Figure 7-1 shows the PDCP counter check process initiated by the SgNB.

Figure 7-1 PDCP counter check process initiated by the SgNB

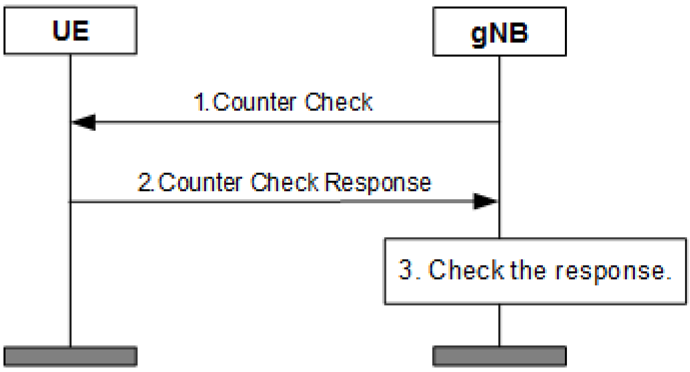


1. The SgNB sends an SgNB Counter Check Request message that contains the DRB IDs and PDCP counters to the MeNB.
2. The MeNB sends a Counter Check message that contains the DRB IDs and PDCP counters to the UE.
3. After receiving the request from the MeNB, the UE checks whether the PDCP counters are consistent.
   - If no inconsistencies are found, the UE returns a Counter Check Response message that does not contain any DRB IDs and PDCP counters to the MeNB.
   - If an inconsistency is found, the UE returns a Counter Check Response message containing the DRB ID and inconsistent PDCP counter to the MeNB.
4. The MeNB receives the response from the UE and checks the response.
   - If the MeNB receives a Counter Check Response message that does not contain any DRB IDs and PDCP counters, the procedure ends.
   - If the MeNB receives a Counter Check Response message that contains a DRB ID and an inconsistent PDCP counter:
     - If the DRB ID returned by the UE does not exist on the MeNB side, the MeNB ignores the DRB ID and performs the check again next time.
     - If the DRB ID returned by the UE exists on the MeNB side, the MeNB checks the value of the **CounterCheckPara**.***CounterCheckUserRelSwitch*** parameter to determine whether to release the corresponding DRB. In NSA networking, the PDCP counter check function can be enabled when the **gNBAirIntfSecParam.*CounterCheckSwitch*** is set to **ENABLE_NOT_RELEASE** or **ENABLE_RELEASE_DRB**. However, the value of this parameter does not determine whether to release the corresponding DRB.

SA Networking

Figure 7-2 shows the PDCP counter check process initiated by the gNodeB.

Figure 7-2 PDCP counter check process initiated by the gNodeB



1. The gNodeB sends a Counter Check message that contains the DRB IDs and PDCP counters to the UE.
2. After receiving the message from the gNodeB, the UE checks whether the PDCP counters on both sides are consistent.
   - If no inconsistencies are found, the UE returns a Counter Check Response message to the gNodeB, without carrying any DRB IDs and PDCP counters.
   - If an inconsistency is found, the UE returns a Counter Check Response message containing the corresponding DRB ID and inconsistent PDCP counter to the gNodeB.
3. The gNodeB receives the response from the UE and checks the response.
   - If the gNodeB receives a Counter Check Response message that does not contain any DRB IDs and PDCP counters, the procedure ends.
   - If the gNodeB receives a Counter Check Response message that contains a DRB ID and an inconsistent PDCP counter:
     - If the DRB ID returned by the UE does not exist on the gNodeB side, the gNodeB ignores the DRB ID and performs the check again next time.
     - If the DRB ID returned by the UE exists on the gNodeB side, the gNodeB determines whether to release the corresponding DRB based on the value of the **gNBAirIntfSecParam.*CounterCheckSwitch*** parameter.
       - If this parameter is set to **ENABLE_NOT_RELEASE**, the corresponding DRB is not released.
       - If this parameter is set to **ENABLE_RELEASE_DRB**, the corresponding DRB is released.

7.2 Network Analysis

7.2.1 Benefits

After this function is enabled, packet loss due to attacks on DRBs between a gNodeB and its UEs can be detected in a timely manner.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

This function has the following impacts on the network:

- If the PDCP counter check function does not find inconsistent PDCP counters, the network is not affected.
- If the PDCP counter check function finds inconsistent PDCP counters, the base station determines whether to release the corresponding DRB based on the configured policy.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Parameter Name	Parameter ID	Setting Notes
Counter Check Switch	**gNBAirIntfSecParam.*CounterCheckSwitch***	It is recommended that this parameter be set to **ENABLE_NOT_RELEASE**, which indicates that the PDCP counter check function is enabled and the corresponding DRB is not released even if inconsistent PDCP counters are found.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the PDCP counter check function without releasing the corresponding DRB
MOD GNBAIRINTFSECPARAM: CounterCheckSwitch=ENABLE_NOT_RELEASE;
```

Deactivation Command Examples

```
//Disabling the PDCP counter check function
MOD GNBAIRINTFSECPARAM: CounterCheckSwitch=DISABLE;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

None

7.4.3 Network Monitoring

- In NSA networking, no network monitoring measure is available after this function is enabled.
- In SA networking, no network monitoring measure is available after this function is enabled.

8 DDoS Attack Detection and Defense over the Air Interface

8.1 Principles

Distributed denial of service (DDoS) attacks are malicious attacks launched by multiple compromised systems on a single target system. A large volume of information flows to the target system, occupying resources of the target system. As a result, the target system is overloaded and refuses to provide services for other UEs. The function of DDoS attack detection and defense over the air interface identifies and isolates abnormal UEs to mitigate the impact of DDoS attacks on the target system. In NSA networking, the function of DDoS attack detection over the air interface is implemented by the eNodeB. For details, see *Radio Security* in *eRAN Feature Documentation*. In SA networking, the function of DDoS attack detection and defense over the air interface is implemented by the gNodeB which identifies and isolates abnormal UEs over the air interface using the following methods:

- During random access of a single UE

  The gNodeB measures the number of RRC connection setup requests, RRC connection reestablishment requests, or RRC connection resume requests sent by a UE within a certain period.

  – If any number exceeds the preset threshold, the gNodeB discards the corresponding requests. Upon determining that a UE is performing a denial of service (DoS) attack, the gNodeB will reject its access for a period of time.
  – If no number exceeds the preset threshold, the gNodeB determines that the UE does not launch a DoS attack and allows the UE to access the network.

- After a single UE enters the RRC_CONNECTED mode

  – After a UE successfully accesses the gNodeB, the gNodeB measures the total number of signaling messages received from the UE on SRB1 and SRB2 within 10s. If the total number exceeds the threshold specified by the **gNBAirIntfSecParam**.***AirIntfSigProtectThld*** parameter, the gNodeB forcibly releases the UE.

- Detection: After a UE successfully accesses the gNodeB, the gNodeB counts the number of malformed packets of the UE at each protocol layer. If the number at any layer exceeds the preset threshold, the gNodeB determines that a malformed packet attack occurs and records this attack in the security log.
- Detection: After a UE successfully accesses the gNodeB, the gNodeB detects abnormal SRs and BSRs of the UE. The gNodeB records security logs when any of the preceding abnormal behaviors occur.
  - Abnormal SRs indicate that the gNodeB detects a consecutively high block error rate (BLER) in the uplink.
  - Abnormal BSRs refer to either of the following situations: (1) The gNodeB detects a high proportion of uplink padding. (2) The value of the following exceeds a preset threshold: Number of UE-reported BSRs carrying control channel data/Number of BSRs with control channel data reported but actually no control channel data transmitted. For details, see the function of stopping uplink scheduling for uplink-abnormal UEs and the method of identifying abnormal BSR UEs in *Uplink Scheduling*.
- Defense against DDoS attacks triggered by SRB scheduling feedback exceptions: This function is controlled by the **SRB_SCH_EXP_SW** option of the **gNBAirIntfSecParam**.***AirIntfAntiDdosSw*** parameter. After this function is enabled, the gNodeB checks whether a UE has consecutive SRB scheduling feedback exceptions before setting up a DRB for the UE. If yes, the gNodeB considers the UE having SRB scheduling feedback exceptions and stops scheduling the UE.

● Detection during random access of multiple UEs
- When multiple UEs send Msg1 messages

  The gNodeB counts the number of invalid preambles received in a cell within certain measurement period. If the number exceeds the preset threshold, the gNodeB determines that a PRACH DoS attack occurs in the cell and records this attack in the security log.
- When multiple UEs send Msg5 messages

  In air interface DDoS attacks that involve multiple UEs sending Msg5 messages, detecting and releasing UEs are based on the function of "defense against DDoS attacks triggered by SRB scheduling feedback exceptions". The function of defense against DDoS attacks triggered by SRB scheduling feedback exceptions is controlled by the **SRB_SCH_EXP_SW** option of the **gNBAirIntfSecParam**.***AirIntfAntiDdosSw*** parameter. After this function is enabled, the gNodeB periodically checks the number of UEs having SRB scheduling feedback exceptions in a cell (the method for identifying UEs having SRB scheduling feedback exceptions is the same as that for the detection and defense procedures of air interface DDoS attack that involves a single UE in connected mode). When the number exceeds a certain threshold, the gNodeB considers that the cell is under a DDoS attack. When the number is below a certain threshold, the cell exits the DDoS attack state. When a cell is under a DDoS attack, the gNodeB performs the following operations on UEs in the cell: The gNodeB releases all UEs having SRB scheduling feedback exceptions. For a UE that subsequently initiates a random access procedure with Msg5 already sent to the gNodeB, the gNodeB checks whether the UE has SRB scheduling feedback exceptions based on the UE's "DedicatedNAS-Message" IE exchange with the CN. If yes, the gNodeB releases the UE.

In the following DDoS attack detection process over the air interface, to detect DDoS attacks from malicious UEs preempting the capacity in terms of the number of UEs, the function of detection of DDoS due to preemption on the number of UEs is introduced. This function is controlled by the **gNBAirIntfSecParam**.***UeNumPreemptDdosChkSw*** parameter. After this function is enabled, the detection process is as follows:

– When multiple UEs send Msg3 messages
  - During cell activation, the gNodeB calculates a threshold: Threshold for the number of emergency UEs released due to timeout = (Maximum number of UEs in the RRC phase in a cell x 0.8/Value of the timer for awaiting UE responses over the air interface) x 5s.
  - The gNodeB measures the number of associated emergency UEs in the RRC phase released due to timeout in a cell during a period of 5s. If this measured number is greater than the preceding threshold, the gNodeB determines that malicious emergency UEs preempt the number of emergency UEs in the cell and records security logs.

  NOTE

  - The value of the timer for awaiting UE responses over the air interface is configured by the **gNBConnStateTimer**.***UuMessageWaitingTimer*** parameter.
  - An associated emergency UE is a UE whose values of "EstablishmentCause" and "ResumeCause" in the RRC connection setup and resume messages of the UE are **emergency**.
  - A UE in the RRC phase refers to a UE in a phase where the base station has sent an Msg4 message to the UE and waits for an Msg5 message from the UE.
  - Release due to timeout: The base station receives an RRC connection setup or resume request from a UE but does not receive any RRC connection setup complete or RRC connection resume complete message from the UE even after the timer for awaiting UE responses over the air interface expires. In this case, the base station will release the UE.

– When multiple UEs send Msg5 messages
  - During cell activation, the gNodeB calculates a threshold: Threshold for the number of UEs in the context phase released due to timeout = (Maximum number of RRC_CONNECTED UEs in a cell x 0.8/Value of the timer for awaiting initial context setup requests) x 5s
  - The gNodeB measures the number of UEs in the context phase released due to timeout in a cell during a period of 5s. If this measured number is greater than the preceding threshold, the gNodeB determines that UEs without contexts preempt the number of UEs in the cell and records security logs.

NOTE

- The value of the timer for awaiting the initial context setup requests is specified by the **gNBConnStateTimer**.***InitContextSetupReqMsgTmr*** parameter.
- A UE in the context phase refers to a UE that has entered the RRC_CONNECTED mode and waits for context setup.
- Release due to timeout: The base station receives an RRC connection setup complete message, an RRC connection reestablishment complete message, or an RRC connection resume complete message from a UE but does not receive any uplink response from the UE after the timer for awaiting initial context setup requests expires. In this case, the base station will release the UE.

- After multiple UEs enter the RRC_CONNECTED mode

  After UEs successfully access the network, the gNodeB sets two thresholds for detection: the threshold for the proportion of emergency call UEs (specified by **gNBAirIntfSecParam**.***EmcUeRatioDetThld***) and the threshold for the proportion of the maximum number of UEs in a cell (fixed value 50%).

  – Periodic detection of the proportion of emergency call UEs in a cell exceeding the threshold

    The gNodeB measures the total number of RRC_CONNECTED UEs and the number of emergency call UEs in a cell every hour. If the proportion of RRC_CONNECTED UEs to the maximum UEs allowed in the cell is greater than or equal to 50%, and the proportion of emergency call UEs to RRC_CONNECTED UEs in the cell is greater than the configured value of **gNBAirIntfSecParam**.***EmcUeRatioDetThld***, the base station determines that the proportion of emergency call UEs in the cell exceeds the related threshold and records the event in the security log.

  – Event-triggered detection when the proportion of emergency call UEs in a cell exceeds the threshold

    When detecting that an emergency call UE preempts resources of a lower-priority UE, the gNodeB immediately triggers a detection on the cell that the emergency call UE accesses. If the proportion of RRC_CONNECTED UEs to the maximum UEs allowed in the cell is greater than or equal to 50%, and the proportion of emergency call UEs to RRC_CONNECTED UEs in the cell is greater than the configured value of **gNBAirIntfSecParam**.***EmcUeRatioDetThld***, the base station determines that the proportion of emergency call UEs in the cell exceeds the related threshold and records the event in the security log.

  NOTE

  - RRC_CONNECTED UEs refer to UEs in the RRC_CONNECTED mode that occupy the capacity in terms of the number of UEs in a cell, regardless of whether the UE context is successfully set up.
  - In event-triggered detection scenarios where the proportion of emergency call UEs in a cell exceeds the threshold, there may be a delay between the time when air interface DDoS attacks preempt resources and the time when security logs are recorded.

- Preemption Ratio Restriction for Multi-UE DDoS Attacks over the Air Interface

  When the number of emergency UEs exceeds a certain threshold, the emergency UEs will preempt UE number resources. Thus, the target system is overloaded and cannot provide services for other UEs. To address this issue, the function of preemption ratio restriction for multi-UE DDoS attacks over the air interface is introduced and it is controlled by the **EMC_UE_PREEMPT_RATIO_LIMIT_SW** option of the **gNBAirIntfSecParam**.***AirIntfAntiDdosSw*** parameter. If this function is enabled and any of the following conditions is met, the base station determines that the UE is initiating DDoS attacks and rejects the UE's access:

- When an emergency UE attempts to access a cell of a gNodeB and the baseband processing unit specifications of the cell are limited, the gNodeB calculates the ratio of the total number of emergency UEs in the cell to the total number of existing UEs in the cell based on the information about UEs served by the baseband processing unit. If the ratio is greater than a specified threshold, the gNodeB rejects the UE's access requests.
- When an emergency UE attempts to access a cell of a gNodeB and the number of admitted UEs in the cell has reached the upper limit, the gNodeB calculates the ratio of the total number of emergency UEs in the cell to the cell-level admission user number threshold for emergency UEs in the cell. If the ratio is greater than a specified threshold, the gNodeB rejects the UE's access requests.
- When an emergency UE attempts to access a gNodeB and the number of admitted UEs of the gNodeB has reached the upper limit, the gNodeB calculates the ratio of the total number of emergency UEs served by the gNodeB to the total number of existing UEs served by the gNodeB. If the ratio is greater than a specified threshold, the gNodeB rejects the UE's access requests.

8.2 Network Analysis

8.2.1 Benefits

The function of DDoS attack detection and defense over the air interface identifies and isolates abnormal UEs to mitigate the impact of DDoS attacks on the target system.

8.2.2 Impacts

The impacts between this function and other functions are described in **8.3.2.3 Function Impacts**.

Preemption ratio restriction for multi-UE DDoS attacks over the air interface: When this function is enabled, if the gNodeB meets both the preemption triggering conditions and the condition that the ratio of emergency UEs exceeds the specified threshold, some emergency UEs are rejected. As a result, the access success rate decreases for emergency call UEs due to preemption rejection, increases for enterprise private line UEs and VoNR UEs, and remains unchanged for UEs of other types.

Defense against DDoS attacks triggered by SRB scheduling feedback exceptions: After this function is enabled, the UEs considered having SRB scheduling feedback exceptions may experience longer access delay and even service drops.

Other functions have no impact on the network.

8.3 Requirements

8.3.1 Licenses

None

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

None

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Air Interface Signaling Protect Threshold	**gNBAirIntfSecParam.*AirIntfSigProtectThld***	None	You are advised to set this parameter to 100 if admitted UEs send a large number of signaling messages.
UE Number Preemption DDoS Check Switch	**gNBAirIntfSecParam.*UeNumPreemptDdosChkSw***	None	It is recommended that this function be enabled when multiple malicious UEs preempt the capacity in terms of the number of UEs on the network during random access of multiple UEs.

Parameter Name	Parameter ID	Option	Setting Notes
Air Interface Anti-DDoS Switch	**gNBAirIntfSecParam**.***AirIntfAntiDdosSw***	SRB_SCH_EXP_SW	It is recommended that this option be selected when the proportion of UEs with SRBs established but DRBs not is greater than 80%. The proportion of UEs with SRBs established but DRBs not is calculated using the formula: **N.RRC.SetupReq.Succ** – **N.PDUSession.Est.Att**)/ **N.RRC.SetupReq.Succ**.
Emergency Call UE Ratio Detection Threshold	**gNBAirIntfSecParam**.***EmcUeRatioDetThld***	None	It is recommended that this parameter be set to 50 if multiple UEs preempt the capacity in terms of the number of UEs after entering the RRC_CONNECTED mode.
Air Interface Anti-DDoS Switch	**gNBAirIntfSecParam**.***AirIntfAntiDdosSw***	EMC_UE_PREEMPT_RATIO_LIMIT_SW	It is recommended that this option be selected when multiple UEs preempt UE number resources after accessing the network.

8.4.1.2 Using MML Commands

The function of DDoS attack detection and defense over the air interface is enabled by default when a single UE initiates DDoS attacks during random access, multiple UEs initiate DDoS attacks by sending Msg1 messages during random access, and a single UE in the RRC_CONNECTED mode initiates malformed packet attacks. MML-based configurations are not required to activate this function. The following provides only MML command examples of air interface signaling protection for a single UE after the UE enters the RRC_CONNECTED mode, detection of DDoS due to preemption on the number of UEs during random access of multiple UEs (only Msg3 and Msg5 messages are involved), and DDoS attack detection over the air interface after multiple UEs enter the RRC_CONNECTED mode.

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the air interface signaling protection function
MOD GNBAIRINTFSECPARAM: AirIntfSigProtectThld=100;
//Enabling detection of DDoS due to preemption on the number of UEs
MOD GNBAIRINTFSECPARAM: UeNumPreemptDdosChkSw=ON;
//(Recommended) Turning on SRB_SCH_EXP_SW when the proportion of UEs with SRBs established but DRBs not is greater than 80%
MOD GNBAIRINTFSECPARAM: AirIntfAntiDdosSw=SRB_SCH_EXP_SW-1;
//Enabling the function of DDoS attack detection over the air interface after multiple UEs enter the RRC_CONNECTED mode
MOD GNBAIRINTFSECPARAM: EmcUeRatioDetThld=50;
//Enabling preemption ratio restriction for multi-UE DDoS attacks over the air interface
MOD GNBAIRINTFSECPARAM: AirIntfAntiDdosSw=EMC_UE_PREEMPT_RATIO_LIMIT_SW-1;
```

Optimization Command Examples

N/A

Deactivation Command Examples

```
//Disabling the air interface signaling protection function
MOD GNBAIRINTFSECPARAM: AirIntfSigProtectThld=65535;
//Disabling detection of DDoS due to preemption on the number of UEs
MOD GNBAIRINTFSECPARAM: UeNumPreemptDdosChkSw=OFF;
//Turning off SRB_SCH_EXP_SW
MOD GNBAIRINTFSECPARAM: AirIntfAntiDdosSw=SRB_SCH_EXP_SW-0;
//Disabling the function of DDoS attack detection over the air interface after multiple UEs enter the RRC_CONNECTED mode
MOD GNBAIRINTFSECPARAM: EmcUeRatioDetThld=255;
//Disabling preemption ratio restriction for multi-UE DDoS attacks over the air interface
MOD GNBAIRINTFSECPARAM: AirIntfAntiDdosSw=EMC_UE_PREEMPT_RATIO_LIMIT_SW-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

- The function of DDoS attack detection and defense over the air interface is enabled by default during random access of a single UE and Msg1 message sending by multiple UEs. You can determine whether DDoS attack detection and defense over the air interface exists based on the counters listed in **Table 8-2**.
  - If any of the counter values is greater than 0, DDoS attack detection and defense over the air interface exists.
  - If all the counter values equal 0, DDoS attack detection and defense over the air interface does not exist.

Table 8-2 Counters related to DDoS attack detection and defense over the air interface during random access

Counter ID	Counter Name
1911823077	**N.RRC.SetupReq.Msg.Disc.ExceedThld**
1911823076	**N.RRC.ResumeReq.Msg.Disc.ExceedThld**
1911823075	**N.RRC.ReEst.Msg.Disc.ExceedThld**

- After a single UE enters the RRC_CONNECTED mode, you can run the **LST GNBAIRINTFSECPARAM** command to check whether DDoS attack detection and defense over the air interface has been enabled. If the value of **Air Interface Signaling Protect Threshold** is not **65535**, this function has been enabled. You can determine whether DDoS attack detection and defense over the air interface exists based on the counter listed in **Table 8-3**.
  - If the counter value is greater than 0, this function has taken effect.
  - If the counter value equals 0:
    - If the value of **Air Interface Signaling Protect Threshold** is not **65535**, DDoS attack detection and defense over the air interface does not exist.
    - If the value of **Air Interface Signaling Protect Threshold** is **65535**, the function of DDoS attack detection and defense over the air interface has not been enabled after the UE enters the RRC_CONNECTED mode.

Table 8-3 Counter related to DDoS attack detection and defense over the air interface after a single UE enters the RRC_CONNECTED mode

Counter ID	Counter Name
1911823074	**N.UECntx.AbnormRel.Sig.ExceedThld**

- During random access of multiple UEs (only Msg3 and Msg5 messages are involved), you can run the **LST GNBAIRINTFSECPARAM** command to query whether the function of detection of DDoS due to preemption on the number of UEs has been enabled. If the value of

gNBAirIntfSecParam.***UeNumPreemptDdosChkSw*** is **ON**, this function has been enabled. In addition, you can check whether DDoS events over the air interface are detected in security logs.

- After multiple UEs enter the RRC_CONNECTED mode, you can run the **LST GNBAIRINTFSECPARAM** command to check whether DDoS attack detection over the air interface has been enabled. If the value of **gNBAirIntfSecParam**.***EmcUeRatioDetThld*** is not **255**, this function has been enabled. In addition, you can check whether DDoS events over the air interface are detected in security logs.
- You can run the **LST GNBAIRINTFSECPARAM** command to query whether the function of preemption ratio restriction for multi-UE DDoS attacks over the air interface has been enabled. If the **EMC_UE_PREEMPT_RATIO_LIMIT_SW** option of the **gNBAirIntfSecParam**.***AirIntfAntiDdosSw*** parameter is selected, this function has been enabled.

8.4.3 Network Monitoring

None

9 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 33.401: "3GPP System Architecture Evolution (SAE); Security architecture"
- 3GPP TS 33.501: "Security architecture and procedures for 5G system"
- 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network(E-UTRAN);X2 application protocol (X2AP)"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification"
- *EPC-based NSA Basics*
- *Uplink Scheduling*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*

